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Project Plan

Smart Wildlife Monitoring System

IoT Project

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1 Introduction

This project implements a smart embedded camera system designed for autonomous wildlife monitoring in remote outdoor environments. The system is optimized for energy-efficient operation and uses a two-stage motion detection architecture that combines low-power hardware sensors with AI-based verification.

The primary target object is the flying squirrel. The system continuously monitors the environment while minimizing average power consumption by activating high-power components only when necessary.

The system is designed to operate autonomously using battery and optional solar power, making it suitable for off-grid deployments.

2 System Overview

The Smart Embedded Camera System consists of:

- OpenMV N6 main processing unit
- OpenMV-compatible camera module
- PIR motion sensor
- Ambient light sensor
- Temperature & humidity sensor
- IR illumination system
- RM520N-GL 5G communication module
- ADC-based current measurement circuit
- 3.7" E-Paper display
- Custom HAT-style PCB
- Battery and optional solar charging system

The system operates using an event-driven state machine architecture.

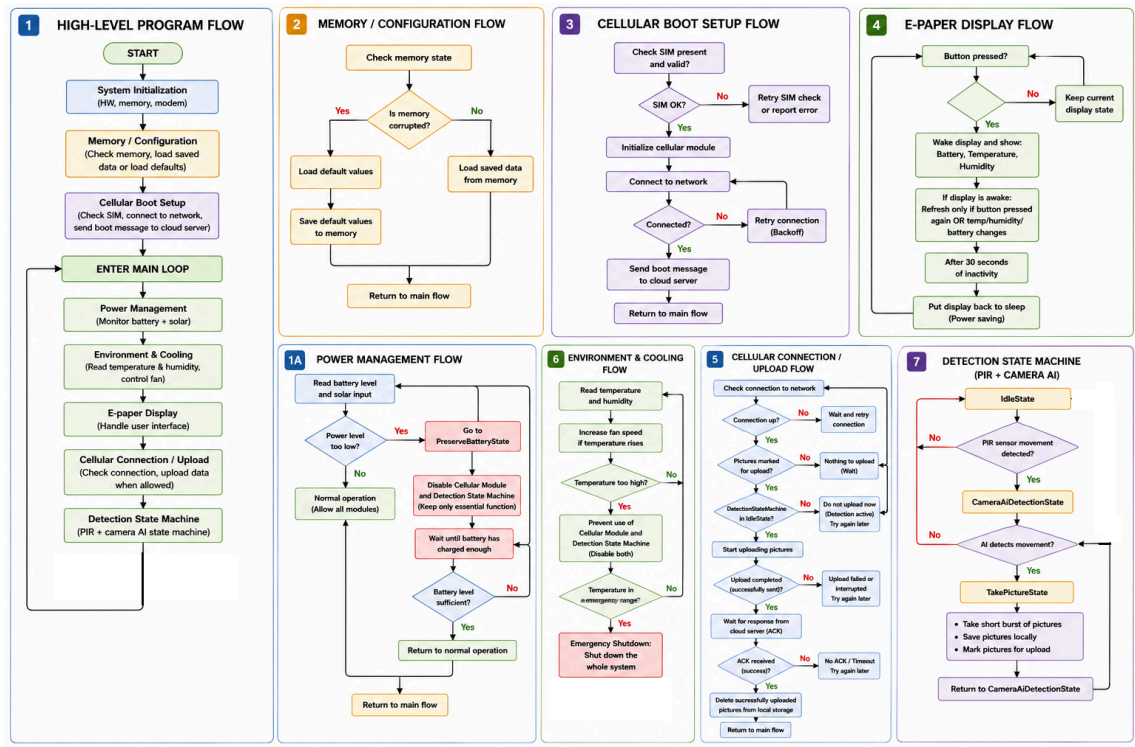


Figure 1. Embedded firmware architecture and state machine flowcharts for the wildlife monitoring system, including power management, memory initialization, cellular communication, environmental monitoring, E-paper display handling, and AI-based motion detection logic.

3 Two-Stage Motion Detection

3.1 Stage 1 - PIR Motion Detection

The PIR sensor continuously monitors infrared movement while the system remains in low-power idle mode. The PIR sensor was selected because of its extremely low power consumption, interrupt-based wake-up capability, and fast response time.

The purpose of the PIR sensor is to detect possible movement without continuously running AI processing or camera capture.

3.2 Stage 2 - AI Verification

The second stage of the motion detection system verifies whether the detected movement is significant enough to justify image capture and cloud transmission.

When PIR movement is detected, the camera system activates and the OpenMV N6 edge AI pipeline starts processing the video stream in real time using the built-in Neural Processing Unit NPU.

The AI system verifies whether the movement is meaningful, filters false triggers, and reduces unnecessary image captures.

The target performance for the embedded AI system is approximately 15–25 FPS processing speed with inference latency below 100 milliseconds. The embedded AI system performs only motion verification, while animal species recognition is handled on the cloud server side.

If movement is confirmed, the system transitions to the image capture state. If movement is not confirmed, the system returns to idle mode.

4 Adaptive Capture Logic

The system uses dynamic capture timing instead of fixed recording periods.

After movement is detected, the camera system remains active for approximately 10–20 seconds. The active duration automatically extends while movement continues. Image capture stops after a configurable inactivity timeout.

This approach reduces average power consumption, improves reliability during wildlife events, and minimizes energy waste caused by false triggers.

5 Day / Night Operation

The system automatically switches between day and night operating modes using an ambient light sensor.

5.1 Day Mode

During day mode, the system uses standard RGB imaging while IR illumination remains disabled.

5.2 Night Mode

During night mode, the NoIR camera module becomes active and the 850 nm IR illumination system is enabled only during motion events.

This approach improves night imaging performance while reducing unnecessary IR power consumption.

6 Cloud Server Logic

After image capture, images are temporarily stored locally before being uploaded to the cloud server using the 5G connection.

The cloud server performs AI image recognition and checks whether target animal species are detected in the uploaded images.

If target species are detected, the images are permanently stored and organized into species-specific folders. The server also stores metadata such as timestamps, detected species, and the number of detected animals.

If target species are not detected, the images are discarded.

The embedded system deletes local image files only after receiving successful ACK confirmation from the cloud server. This prevents data loss during transmission failures.

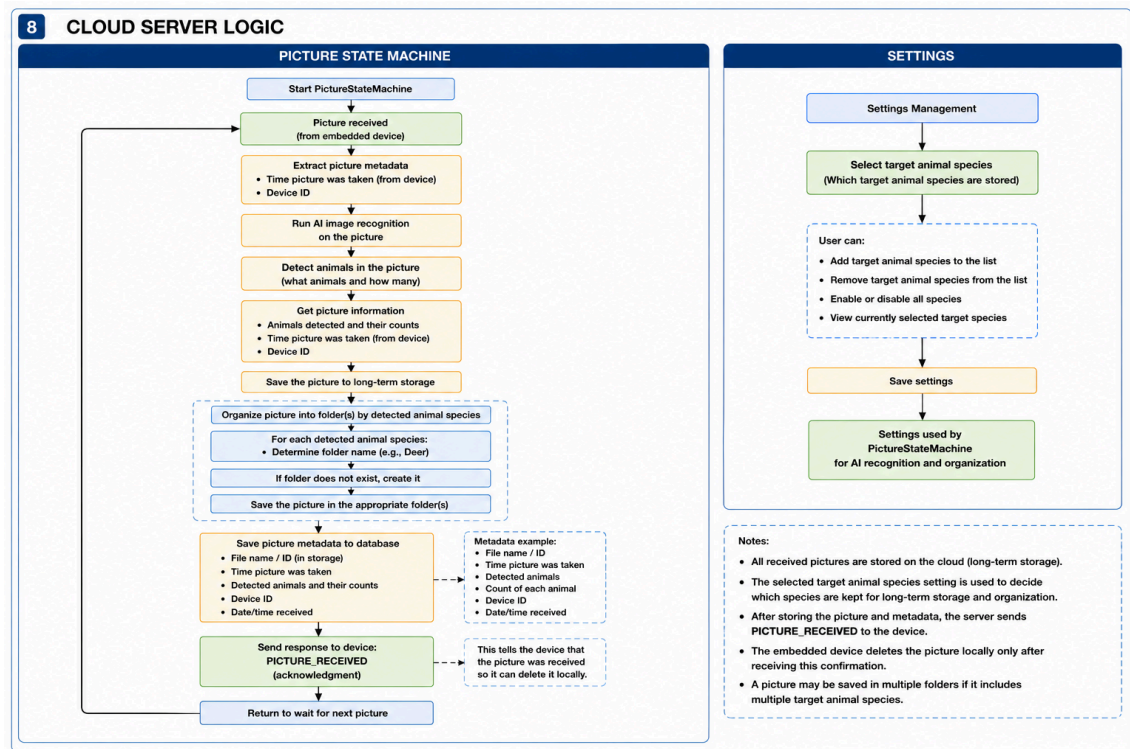


Figure 2. Cloud server architecture and AI image processing workflow used for animal recognition, metadata handling, long-term storage management, and embedded device acknowledgement.

7 Power Management & Energy Supply

The power management system is designed to support long autonomous operation in remote outdoor environments.

The maximum current consumption is estimated by summing the worst-case current draw of all active components. Average power consumption is estimated using expected active and standby operating times. These estimates are used to approximate battery operating time.

7.1 Power Strategy

During idle mode, only the PIR sensor and essential environmental sensors remain active. High-power components such as the camera, AI processing pipeline, IR illumination system, and 5G modem remain disabled.

During active mode, the camera system, AI processing, and wireless communication modules become active only after verified motion events.

7.2 Energy Source

The primary energy source is a Renogy 12V 300Ah LiFePO₄ battery with an approximate capacity of 3840 Wh. The 5G communication subsystem requires a stable 5 V power supply because transmission bursts can generate high peak current consumption.

The system also supports optional solar charging.

The LTE transmission system is expected to consume approximately 7–12 W during active data transmission.

8 Hardware Components

This chapter presents the major hardware components used in the Smart Embedded Camera System.

8.1 Processing Unit

The main processing unit of the system is the OpenMV N6 machine vision module based on the STM32N657 microcontroller.

The OpenMV N6 includes a Cortex-M55 processor, built-in Neural Processing Unit NPU, embedded AI acceleration support, GPIO interfaces, USB-C connectivity, wireless connectivity support, and microSD storage support.

The processing unit is responsible for embedded AI motion verification, state machine control, sensor management, communication handling, and image processing.

8.2 Camera Module

The system uses an OpenMV-compatible camera module connected to the OpenMV N6 platform.

The camera subsystem supports embedded image capture, real-time AI-assisted motion verification, and infrared-compatible night imaging.

The camera system is optimized for low-power embedded operation and wildlife monitoring applications.

8.3 Motion Detection

The motion detection system uses a Grove PIR Motion Sensor.

The PIR sensor was selected because of its ultra-low-power operation and GPIO interrupt support.

8.4 Environmental Sensors

The environmental monitoring system includes an ambient light sensor and a temperature and humidity sensor.

The ambient light sensor is used for automatic day and night mode switching.

The temperature and humidity sensor is used for enclosure monitoring, condensation prevention, and overheating protection.

Both sensors communicate using the I²C interface.

8.5 IR Illumination System

The IR illumination system uses a Waveshare light-sensing infrared LED board operating at 850 nm wavelength.

The IR illumination system activates only during motion events. MOSFET-controlled switching is used to support external high-current operation.

8.6 Communication Module

The communication system uses the RM520N-GL 5G HAT.

The module communicates using a PCIe Gen2 ×1 interface and supports high-speed image upload through existing cellular infrastructure.

The expected sustained upload speed is approximately 5–20 Mbps.

8.7 Power Monitoring

Power monitoring is implemented using an ADC-based current measurement circuit.

Current is measured using a low-value shunt resistor connected in series with the system power line. The voltage drop across the shunt resistor is measured using an ADC input and the current is calculated using Ohm's law.

The measured current can be used to estimate system power consumption, battery usage, and operating time during active and low-power modes.

8.8 User interface

The user interface uses a 3.7-inch E-Paper display with 480 × 280 resolution and SPI communication.

The display shows important system information such as battery level, storage usage, network status, temperature, motion detection state, and capture count.

The E-Paper display was selected because it consumes very little power and retains its image without continuous power.

9 Local Configuration Options

The local maintenance interface allows configuration of essential embedded system settings directly from the docking keypad and E-Paper display.

The system stores configuration settings locally on the microSD card using configuration files. The microSD card is also used for temporary image buffering and offline storage during communication failures.

Device Settings

- Device name
- Administrator password
- System reboot

Server Settings

- Server address
- Username
- Password
- Connection test

Network Settings

- Cellular APN settings
- Network connection test

Motion Detection Settings

- Motion sensitivity
- Capture interval

Camera Settings

- Night mode ON/OFF
- IR brightness

Basic Status Information

- Battery level
- Cellular signal strength
- Storage usage

10 Interfaces and Technical Design

This chapter presents the communication interfaces used in the Smart Embedded Camera System.

10.1 GPIO

GPIO connections are used for the PIR sensor, IR emitter control, and interrupt handling.

GPIO was selected because it provides simple digital signaling with very low power consumption.

10.2 I²C

The I²C interface is used for the ambient light sensor, temperature and humidity sensor.

I²C was selected because it supports shared low-speed communication using minimal wiring.

10.3 CSI-2

The camera module is connected directly to the OpenMV N6 platform using the supported embedded camera interface.

The interface supports low-latency image transfer for embedded AI processing and motion verification.

10.4 PCIe

The PCIe interface is used for the RM520N-GL 5G communication module.

PCIe was selected because it provides significantly higher bandwidth than UART-based communication.

10.5 SPI

The SPI interface is used for the E-Paper display.

SPI was selected because it provides efficient low-power display communication.

10.6 ADC

The ADC interface is used for current measurement and battery monitoring.

Current measurement is implemented using a shunt resistor connected in series with the power line. The ADC measures the voltage drop across the resistor and the embedded system calculates the current using Ohm's law.

11 PCB and Physical Design

This chapter presents the PCB implementation and physical enclosure design.

11.1 PCB Design

The system uses a custom Raspberry Pi 5 compatible HAT-style PCB with approximate dimensions of 65 x 56 mm. The PCB is compatible with the OpenMV N6 platform through supported communication interfaces.

The PCB uses a two-layer design and includes GPIO headers, CSI routing, shared I²C communication, PCIe modem routing, power regulation circuitry, fan connectors, and sensor connectors.

The routing process was completed manually and the B.Cu layer was used as a ground plane to reduce electrical noise and improve signal reliability.

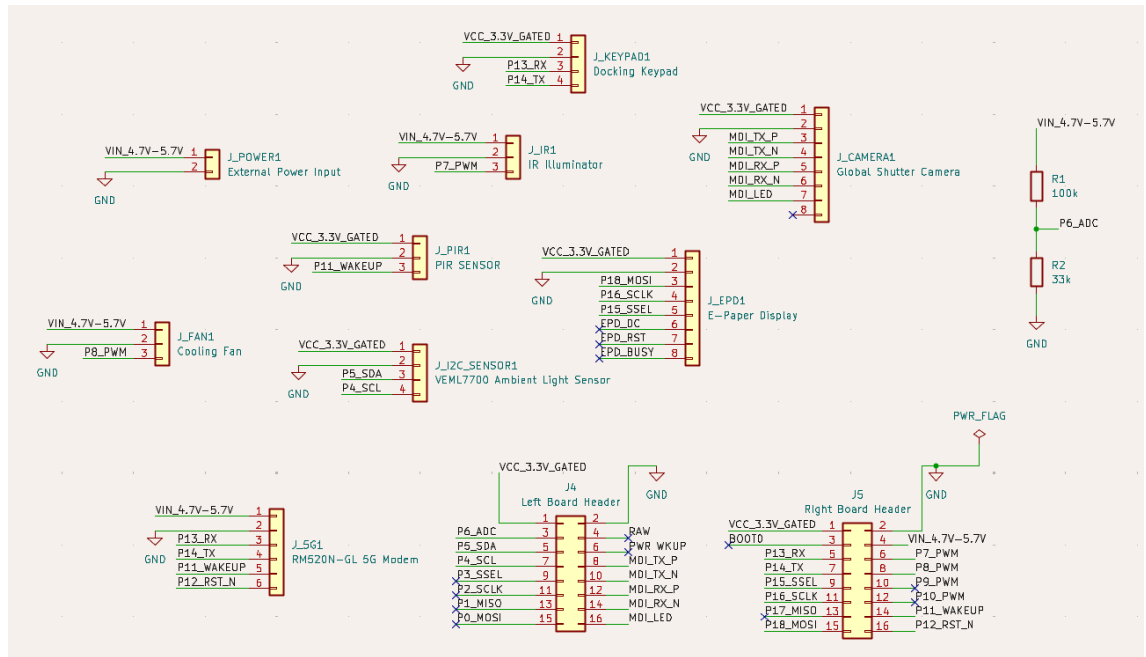


Figure 3. High-level schematic implementation of the embedded wildlife monitoring system.

11.2 Design Verification

The PCB design was verified using Design Rule Check (DRC), short-circuit verification, and unconnected net verification.

Manufacturing outputs include copper layers, silkscreen layers, drill files, board outlines, and Gerber files.

The final PCB design is suitable for real-world manufacturing.

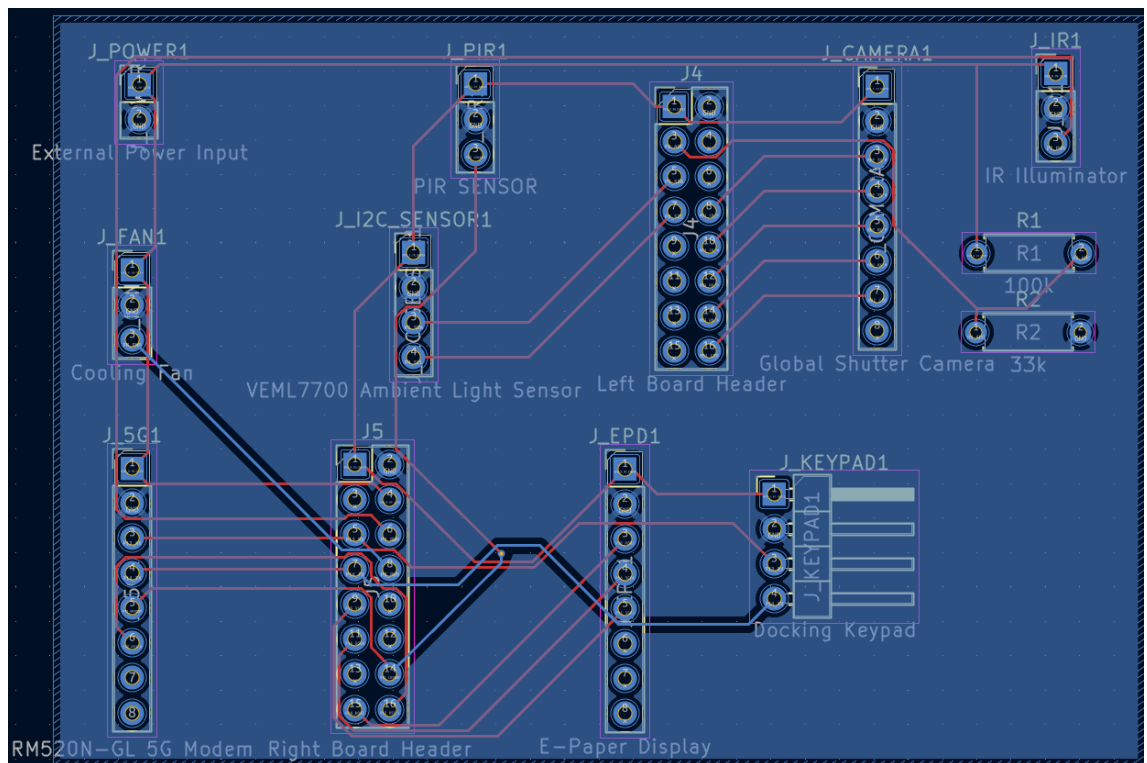


Figure 4. PCB trace routing and component placement of the custom OpenMV N6 expansion board.

11.3 Physical Enclosure

The approximate enclosure dimensions are 110 × 90 × 60 mm.

The enclosure is designed for outdoor deployment and includes airflow support, environmental protection, and sensor protection.

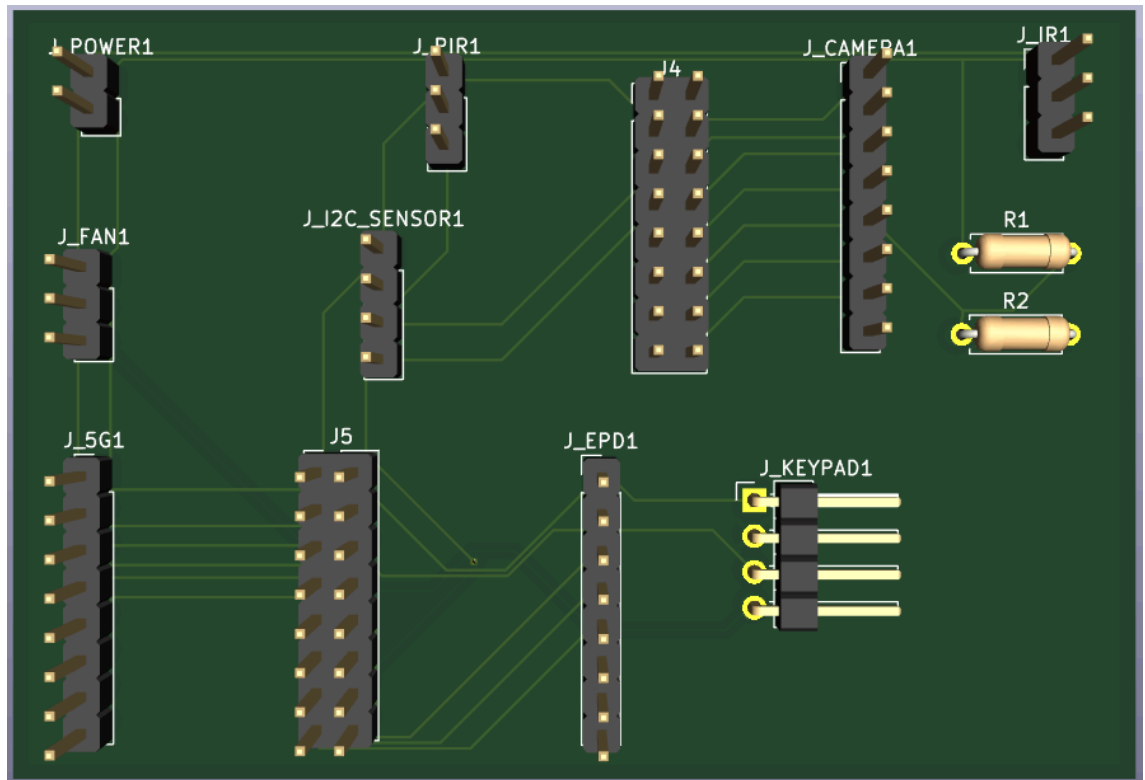


Figure 5. 3D visualization of the custom embedded system PCB showing component placement and connector layout

12 Technical Specifications

Category	Specification
Processing	OpenMV N6
AI Performance	TensorFlow Lite, 15-25 FPS
AI Latency	<100 ms
Motion Detection	PIR + AI verification
PIR Latency	<= 10 ms
Video	1280x800 @ 120 FPS
Images	1 MP global shutter imaging
Night Vision	NoIR + 850 nm IR
Communication	RM520-GL 5G

Uplink Speed	5-20 Mbps
Energy Supply	12V 300Ah LiFePO4
PCB Size	~65 x 56 mm
Enclosure Size	~110 x 90 x 60 mm
Display	3.7-inch E-paper SPI

13 Future Improvements

Possible future improvements include improved enclosure design, more advanced power management, additional environmental sensors, long-term field testing, further AI optimization, solar charging optimization, and expanded cloud analytics.

14 Conclusion

The Smart Embedded Camera System combines embedded systems, AI-assisted motion detection, wireless communication, and energy-efficient hardware design into a single autonomous wildlife monitoring platform.

The project demonstrates the integration of embedded systems, edge AI processing, PCB design, cloud-integrated image processing, and autonomous off-grid operation.

The system is designed to be scalable, field-deployable, and suitable for real-world environmental monitoring applications.